

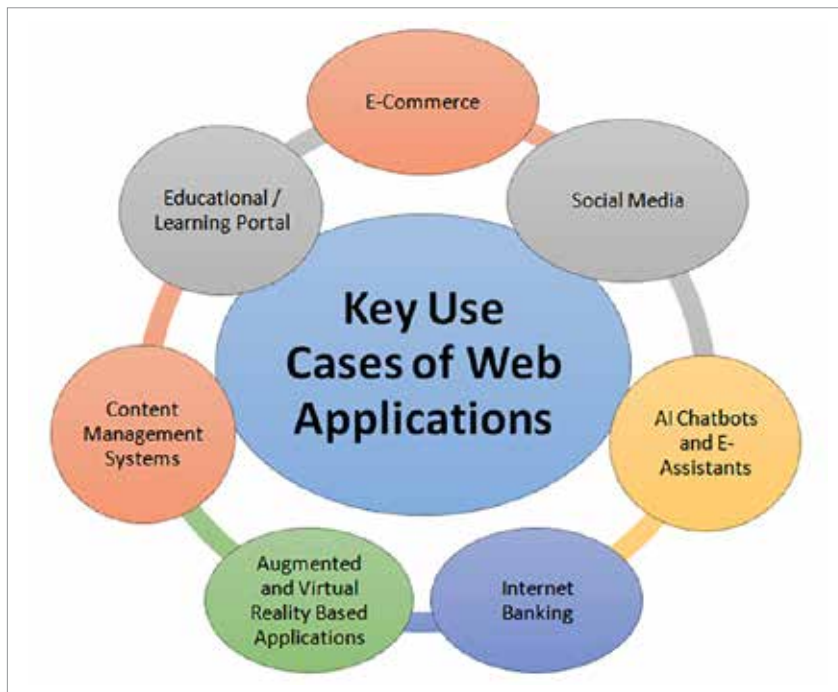


Figure 1: Key use cases of web development

E-commerce: Web development is pivotal in creating platforms that facilitate online transactions, product browsing and customer service. Web applications like Amazon exemplify how sophisticated web development can drive commercial success by providing seamless, user-friendly interfaces coupled with robust backend systems that manage millions of transactions daily. AI is redefining the shopping experience by enabling personalised recommendations, automated customer service and efficient inventory management. This integration of AI into e-commerce platforms is a key factor driving the market growth for AI-based web and mobile applications.

Healthcare: The healthcare industry is witnessing a paradigm shift through the implementation of telemedicine and electronic health records (EHR) systems, which rely heavily on secure, robust web applications. These systems enable remote consultations, real-time patient monitoring and efficient management of patient data, thereby improving the quality of care and patient outcomes.

The healthcare sector is experiencing a transformative impact from AI-based web and mobile applications. AI-powered diagnostic tools and health monitoring apps are revolutionising patient care by enabling early detection of diseases, personalised treatment plans and remote patient monitoring.

Social networks: In the realm of social networking, platforms such as Facebook and Twitter rely heavily on complex web development to manage vast amounts of user-generated content, real-time communication and personalised user experiences.

Finance: The financial services industry is leveraging AI to enhance fraud detection, risk management and customer service. AI algorithms analyse transaction patterns and detect anomalies in real-time, thereby preventing fraudulent activities and ensuring secure transactions. In addition, AI-powered robo-advisors provide personalised investment advice based on individual financial goals and risk profiles, democratising access to financial planning services.

Education and learning portals:

In the educational sector, web development has revolutionised the way knowledge is disseminated and consumed. Online learning platforms like Coursera, Udemy, Khan Academy and others employ advanced web technologies to deliver interactive courses, video lectures and real-time assessments, making education more accessible and flexible.

E-governance: Government and public services also benefit from web development by offering online portals where citizens can access information, submit applications and interact with government services, thus enhancing efficiency and transparency.

Market size of AI-based web and mobile applications

The market size of AI-based web and mobile applications has witnessed exponential growth in recent years, driven by rapid advancements in artificial intelligence and its integration into various facets of digital technology. As AI technologies continue to mature, their applications in automating processes, analysing vast datasets and providing predictive insights are becoming increasingly ubiquitous, thereby expanding the market landscape.

According to recent industry reports, the global market for AI-based applications is poised to reach unprecedented valuations. The market was valued at approximately US\$ 40 billion in 2020 and is projected to grow at a compound annual growth rate (CAGR) of over 40% from 2021 to 2028. This remarkable growth trajectory can be attributed to the escalating investment in AI research and development by major tech companies, startups and government bodies worldwide.

One of the key drivers of this market expansion is the widespread implementation of AI in customer service applications. Chatbots and